

Biotensegrity in sport and movement

John Sharkey

*"We snatch in vain at Nature's veil
She is mysterious in broad daylight,
No screws or levers can compel her to reveal
The secrets she has hidden from our sight."*

Faust by Johann Wolfgang von Goethe

Introduction: tensegrity versus biotensegrity

Dr Donald E. Ingber outlined his findings from 30 years of research on tensegrity and mechanotransduction at the 1st International Fascia Research Congress (IFRC), at Harvard Medical School Conference Center. Ingber revealed that "[M]olecules, cells, tissues, organs, and our entire bodies use 'tensegrity' architecture to mechanically stabilize their shape, and to seamlessly integrate structure and function at all size scales."

However, one could ask the question "What consequences does this have for individuals working in sport and movement"? The answer is clear. Scientific evidence was finally established supporting the notion that mechanical forces produce change in our biochemistry, gene expression, metabolism and our overall physiology (Ingber, 2008). No longer would genes, chemicals and DNA hold exclusive domain over our internal physiology.

Mechanical properties, behavior and movement were now understood to be serious contenders worthy of consideration in the scientific investigation concerning the health of our cells. Pioneering research from Ingber led to

a proposition to replace accepted engineering principles that were firmly established as the methodology used to solve medical problems. The model of tensegrity was put forward as a means to enlighten bioengineering.

This, in effect, represented a role reversal, whereby nature would inform engineering (bioinspired) rather than non-biological engineering-inspired solutions for interventions such as hip replacements or drug delivery applications. The implications of tensegrity in sport and movement are now beginning to be appreciated.

Orthopedic surgeon Stephen Levin created the term "biotensegrity", a word comprising three parts, bio, tension and integrity. An examination and appreciation of Buckminster Fuller's ideas and Levin's unique interpretation of Kenneth Snelson's sculptures gave rise to this new concept. It became apparent that Newtonian, Hookean and linear mechanical properties are the basis for the building of all things non-biological (Levin, 1995). Buckminster Fuller (1895–1983), an architect of fame in the mid-twentieth century, coined the term "tensegrity" by blending the words tension and integrity to create a portmanteau of tensegrity. It was a genius move, encapsulating the essence of what makes tensegrity and biotensegrity so distinctive.

Fuller would later describe tensegrities as "islands of compression in an ocean of tension". In that simple phrase exists the appealing nature of tensegrity (and hence biotensegrity) for the sport and movement specialist.

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As a metaphor (not reality for a biotensegrity), one can instantly imagine the “ocean of tension” representing the fasciae while “islands of compression” could be represented by bones (i.e. the skeletal components invested in the fascia matrix). The fascial oceans become seas, lakes, rivers, streams and brooks embracing and becoming the sandy shores of the osseofascial tissues (Sharkey, 2018a). This description supports the more recently accepted image of a continuous tissue, ubiquitous in nature, connecting left to right, front to back, top to bottom, embracing and permeating the entire body. The visceral organs integrate structurally and physiologically into this system.

The upper and lower limbs are semi-rigid, non-linear bony segments. These segments are interconnected by non-linear connectors. These include cartilage, joint capsules and ligaments with an integrated non-linear, active motor system including the muscles, tendons and fascia. There are no limb segment boundaries. The smaller bones and joints of the hands and feet fully integrate into the biotensegrity model. The spine is a tensegrity turret that integrates with the limbs, head and tail, and also to the visceral system.

A change of tension anywhere within the system is instantly signaled to everywhere else in the body both mechanically and chemically. There is a total body response by mechanical transduction. Transduction involves a biological membrane receiving and transducing or changing a mechanical force (including light and sound) into a chemical signal or neural impulse.

However, there is a bigger, more accurate, picture. At first glance tensegrity seems like an easy model to grasp, especially as many authors use simple tensegrity models to help explain biotensegrity (Figure 11.1). It is important that the reader recognizes the varied limitations of a tensegrity model as this will help to avoid

over-simplification. The tensegrity icosahedron reflects the natural process by which forces orchestrated by the environment dictate form and function. The tensegrity icosahedron is a force vector model that helps to explain what seems to be unexplainable.

Fuller first realized the significance of the tensegrity model, as a new structural design principle, in 1948 when examining a bespoke model constructed by one of his students, the artist Kenneth Snelson (1927–2016). Fuller later described Snelson’s “tensegrity” model by, as previously mentioned, blending the words tension and integrity as a syntactic. A move that encapsulated the essence of what makes tensegrity so distinctive and so immediately understood at the basic level. The tensegrity has no breath or conscience and no life force. It is, in essence a model which we can use to understand the architecture of living structures from cells to whole organ systems.

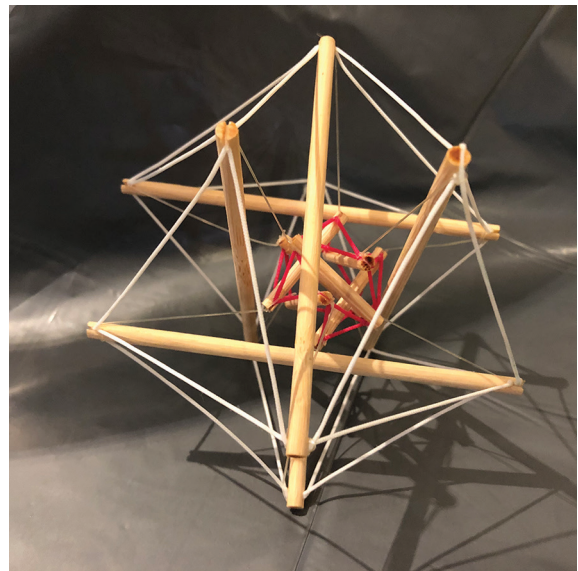


Figure 11.1 Tensegrity with a tensegrity

Model by Graham Scarr.

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The tensegrity model has generated substantial international excitement. Its relevance in aiding sport scientists and movement practitioners to better understand the human body is gaining traction. Tensegrities contain an intriguing simplicity that, in keeping with the laws of nature, is comparable with all living structures: efficiency and the minimization of energy (Avison, 2014). It is fortunate that research pertaining to biotensegrity and fascia has reached new levels of interest almost in parallel. While the fascial system can be viewed as the fabric of continuity and communication, biotensegrity can be seen as the model for explaining the architecture underpinning continuity and communication.

Holding a tensegrity (Figure 11.1), one immediately notices the following:

1. The tensegrity structure is self-stressed and avoids collapse due to the opposing effects of its continuous elasticated member and its discontinuous compressional struts (Figure 11.1). (Note that in bodywork and movement science the opposite of collapsing would be expansive.)
2. Attempts to compress the entire structure will result in the tensegrity collapsing. It will regain its original shape and integrity once the external force is removed, returning to a position of stability and equilibrium.
3. Movement at any point, or location, of the tensegrity results in those forces being distributed throughout the entire tensegrity structure.
4. The struts (or bones) do not touch.
5. There are no joints (i.e. pin joints).
6. There is no friction (i.e. no sliding).
7. There are no bending or shear moments.
8. Attempts to pull on any aspect of the tensegrity causes an increase in “stiffness”

and expansion throughout the entire structure.

9. Tensegrity is light in weight, strong and very resilient and can change shape with minimal demand on energy.
10. Tensegrities exhibit linear properties while a biotensegrity is non-linear.

What is less immediately noticeable is that tensegrities or biotensegrities do not need gravity (Eyskens, 2020). Biotensegrities can operate in any position on earth or on the space station when floating (e.g. hand-stands, cartwheels, side-lying, summersaults, high board diving and gymnastics).

Fuller had long been searching for architectural solutions in nature comparable to tensegrity because it is structurally and functionally efficient, lightweight, dynamic, and adjusts itself to its surroundings. He observed that all natural forms were the consequence of matter being dictated by forces. He proposed them as finite energy systems consisting of the forces of tension and compression acting together (Scar, 2018).

Fuller began working on a theory that respected the shape of space as central to the dynamics of nature. He even gave it a name: synergetics, the study of nature's coordinate system. Fuller used geometry to exhibit a dynamic architecture that was based on first principles (Fuller, 1975).

Nature's way—C'mon let's twist again

René Descartes (1596–1650), a French philosopher, mathematician, and scientist, put forward the model that living anatomy reflects the structure expected of an automated machine. In that regard the skeletal system was seen as separate to muscles and internal organs. A bone was considered to be a separate structure from all other bones and so the foundation was set for the model of parts rather than a model of the

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whole. Pin joints were the order of the day, leading to the modern day vision where surgery for hip, knee and vertebral wear and tear (or pathology) provides the evidence that we can replace these structures with materials that, while not the same, were considered mechanically similar to a knee and a hip.

In 1794, when visiting the University of Jena, Alexander Von Humboldt was working on experiments with his close friend, Johann Wolfgang von Goethe, Germany's beloved poet and author on the topic of comparative anatomy. Johann Friedrich Blumenbach had been Humboldt's professor when a student at university. Blumenbach developed theories concerning forces that shaped organisms, which he called "formative drive and vital forces".

Goethe and Humboldt collaborated so as to integrate Blumenbach's ideas into their own theories. Goethe wrote, "The snake, for example, has an endlessly long neck because 'neither matter nor force' had been wasted on arms or legs. By contrast, the lizard has a shorter neck because it has no legs, while the frog has an even shorter neck because its legs are longer."

Goethe went on to explain his belief that, contrary to Descartes's theory, animals were machines made up of autonomous parts, that a living organism was in effect a functional unified whole. One of the epistemological principles of biotensegrity is the explicit nature of our continuity. Contrary to Descartes, Goethe explained that while a machine could be dismantled and assembled again, the parts of a living organism work only in relation to each other. In a mechanical system the parts shape the whole, while in an organic system the whole shapes the parts. Humboldt interpreted the natural world as a unified whole that is animated by interactive forces (Wulf, 2016). Great minds of antiquity were

coming to the tensegral reality of nature, forces, resulting proportions, symmetry, patterns, spirals, curves, heterarchy and hierarchy.

In his seminal book, *On Growth and Form*, the Scottish mathematical biologist D'arcy Wentworth Thompson described how "the forms of living things and the parts of living things can be explained by physical considerations and to realise that in general no organic forms exist save as are in conformity with physical and mathematical laws" (Thompson, 1917). In his recent, thought-provoking book, *Seeking Symmetry*, Dr Niall Galloway explains that "our human bodies are one of those living things, also 'so many portions of matter', and conformed in accordance with those same general physical laws" (Galloway and McLaren, 2018).

Today, we are coming to terms with the biotensegral nature of our continuity and the force vectors that are dictating morphological outcomes, metabolism and the physiology of all bio-systems, a reality pointed to by Goethe, Humboldt and D'arcy Thompson, among others.

In keeping with these dictates, nature has determined that everything from the cosmos to microbes is assembled upon a self-generated helical architecture. Human movement is predicated upon closed kinematic chains (Levin et al., 2017), reflecting the fractal nature of our cytoskeleton from its molecular to tissue-level organization, demonstrating the continuous nature of our heterarchy. Open kinematic chain mechanisms exhaust energy and are demanding, unidirectional, disjointed structures that are linear-based. According to Scarr (2018) the geometry of motion expressed in kinematic terms provides explanations concerning the forces of compression and tension continuously linked in a manner that alters the shape of the entire system (i.e close kinematic system).

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The helical arrangement of monomers made of H- and G-proteins constitute contractile actin proteins. The more complex collagen fibers are wrapped up in a chain of left- and right-handed helices forming a cross-helical arrangement (chirality) and circumferential wrap architectural configuration. This creates shape-changing tubes within tubes. An example of the helical nature of our form is the so-called ventricular myocardial band concept of the Spanish cardiologist, Torrent Guasp, although the true anatomy of the heart is currently under debate. The helical ventricular myocardial band is an attempt to explain the heart's dynamic actions of "narrowing, shortening, lengthening, widening, twisting, and uncoiling" (Buckberg et al., 2018).

Same, same but different

Bone brittleness is approximately the same in marsupials as in a rhinoceros because the stiffness and strength of bones is approximately the same in all animals. Based on classical biomechanics, animals bigger than lions, such as horses, should break and fracture their bones with a single leap when running or jumping on their slim limbs. Based on linear mechanical laws, including Galileo's square-cube law (which are the foundation of biomechanical models), animal mass must be cubed as their surface area is squared. This should mean that animals as big as a rhinoceros will collapse under their own weight.

Working elastically at strains that are $10 \times 10 \times 10$ times greater than strains that ordinary technological solids can withstand, demonstrates that biological tissues behave differently when compared to non-biological materials. If this was not the case, Levin (1982) informs us that "the skull should explode with each heart beat due to the blood vessels expanding and crowding

out the brain. As they reach fullness the urinary bladder should thin and burst. During pregnancy the uterus should burst with the contractions of delivery."

Not exclusively mechanical, physiologic processes would be inconsistent with linear physics. Pressure within a balloon decreases as a balloon empties. Humans are omnidirectional in order that the tension elements function at all times in tension, regardless of the direction of applied force. In turn, the compression elements in biological structures "float" in a tension network. In a continuous compression model, "the shoulder has little relationship to the foot, the sub-occipital structures work autonomously with no concern for the sacrum" (Sharkey, 2018b).

The basic force-generating components for a tensegrity structure include compression and tension combining to distribute and balance mechanical stresses. Compression and tension are often described as "invisible forces" (Scarr, 2018).

It has been proposed that tension and compression represent the common, or global, language of the human body (Sharkey 2018b). At the cellular level this language needs to be translated into metabolism and physiology otherwise referred to as mechanotransduction.

Simplicity and efficiency are the building blocks for making the most complex organic or inorganic materials or organisms. This language can change depending on the scale; for example, at the molecular level, tension could be replaced with "repulsion" while compression could be replaced with "attraction".

Although not a perfect fit, we can, as a language of convenience, look at the struts of a basic tensegrity as representing our bones. The cables, be they rubber or metal, would represent the

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soft tissue tensional members, including muscle fibers, tendons, ligaments, cartilage, menisci, incompressible organs, nerves, vascular, lymphatic and integument.

According to Scarr (2018), “[A] strut should be considered as any structural entity that maintains a separation between one point and another - a bone, muscle, blood vessel, fascial compartment and even a single cell - and which collectively become the volumetric ‘space generators’ of the body.”

Building a biotensegrity

To construct a biologic organism on the principles of biotensegrity, it must first have the potential to build itself. The structure would be one integrated tensegrity truss that evolved from infinitely smaller trusses, which could be both structurally independent and interdependent at the same time (Figure 11.2). Ingber (2000) described this truss as the icosahedron. Biologic tissues, such as muscles and other soft tissue, have non-linear stress/strain curves.

In classical biomechanics, the original principle that muscles, fascia and any biologic material can be compelled by the rules of hard matter physics is faulty according to the scientific evidence (Ingber, 2008) emanating from laboratories worldwide involved in investigating and researching condensed soft matter.

Ligaments and fascia, bones and cartilage would do little to support our upright forms if not for the collective activity of an integrated osseofascial/myofascial system. Pre-stress is a vital aspect of biotensegrity, with a level of tension present always within the myofascial (or neuromuscular) system. Unfortunately, scientific endeavors investigating muscle tone seldom, if ever, take bone into consideration. Malleable, bending and rotating are words we can use to

describe the function of bones. Concerning the osseofascial system, this point is crucial, as the bony points researchers use to calculate the contractile elements and the elastic elements (i.e. tendonous) would obviously move. Bony points of reference, in this regard, would not be ideal for such measurements, and any conclusions derived would fall short of the robust reality required for gold standard, peer-reviewed research.

Resting tone

There are no resting muscles in living anatomy and physiology. The non-linearity of collagen does not allow for zero tension in human contractile tissue. In other words, contractile tissue never rests. We must also consider Lombard's paradox (Andrews, 1987), which is present in all two joint muscle pairings throughout the body and in closed kinematic chains. Based on classical neuroanatomical laws, it may seem ridiculous to suggest that the upper portion of the hamstring muscle group contracts while the bottom half of the quadriceps contracts simultaneously (i.e. a co-contraction). However, the paradox becomes clear when we consider the action of sitting to standing (Figure 11.2).

When sitting, rectus femoris, the largest of the quadriceps muscle group, is hypothetically in a shortened phase as the hip is in a flexion phase. Rectus femoris runs over the anterior aspect of the flexed knee enveloping the patella and associating with the tibial tuberosity by means of the patellar ligament. It is, therefore, hypothetically lengthened at the knee when sitting but shortened at the hip. Meanwhile, the hamstrings are lengthened at the hip and shortened at the posterior knee. When we move from sitting to standing our hip must extend and our knee must also extend. Rectus femoris therefore hypothetically elongates at the hip and shortens at the knee.

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Figure 11.2 Sitting to standing.

Co-contraction of quadriceps and hamstrings—Lombard's paradox. The arrows show direction of movement relative to the joints.

Dr John Cleland was Professor of Anatomy and Physiology at Queen's College, Galway, Ireland, from 1863 until 1877. He stated that “I have made the measurements of the distance from the superior to the inferior attachment of the gastrocnemius muscle, when both knee and ankle were completely flexed, and when they were both completely extended, and have found that the distance remains unchanged. If we sink upon bended knees, flexing the limbs completely and remaining balanced on the toes then to rise to our full height on tip-toe, the length of the

gastrocnemius remains unchanged in the movement” (Cleland, 1870).

From the biotensegrity viewpoint these are classical examples of how tissues neither lengthen nor shorten but rather shape change throughout range of motion, similar to prismatic architectural materials changing shape and reconfiguring their morphology without any individual part necessarily becoming shorter, longer, wider or narrower (Purslow, 2010). This calls into question the idea of “agonist” and “antagonist” from the classical neurological description (i.e. when an agonist contracts its antagonist relaxes). This simple example of muscular co-contraction does not include consideration of the pelvis or the tibia. Considering that the pelvis is not fixed in space, it would therefore seem reasonable to suggest that current research is working off faulty models at best. So how does this paradox allow for efficient movement?

Cadaveric investigations concerning the role of fasciae in force transmission and mechanical transduction have motivated further investigations into the possibility of this paradoxical muscle-fascial activity (Loukas et al., 2007). Observational dissection investigation on Thiel soft fixed cadaveric specimens has provided insights concerning the non-linear load distributing myofascial synergistic mechanisms of force vector dispersing of loads and conversion to movement. Fascial septal tissue provides connections, allowing force vectors in a hierarchical helical arrangement from anterior inferior (in both medial and lateral directions) to superior posterior, or vice versa (Buckberg and Gerard, 2002).

Fractals

In 2012, scientists discovered the so-called “God particle”, otherwise known as the Higgs Boson.

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Now that we know the Higgs Boson theoretically exists, we can state that it must be made of something smaller. Everything we are made of can be seen, hypothetically, as either a cable or a strut, with each cell being made up of smaller cables and struts of different size scales (i.e. of a fractal nature), with “the relations among their parts dynamic, contextual and interdependent” (Saetzler et al., 2011). Biotensegrities are truly bottom-up and top-down structures whose parts are related to and interdependent upon the whole (Figure 11.3).

The whole is related to and interdependent upon the parts resulting in emergent properties that would not be obvious when investigating only a part (Ritter and Bailey, 1928).

Levin (2018) states that “[T]ensegrity is a multi-scale, heterarchical structure and that any events at one level will, necessarily, have a structural

affect on all other component parts of the organism. This also applies to the dynamical mechanical events occurring within the organism, as the kinematics and kinetics of the organism are intertwined within the tensegrity structure.”

A “structural effect” simply means that movement at the foot will have, for example, structural impact elsewhere in the body, perhaps at the contralateral shoulder or ipsilateral hip (or vice versa). While one can argue correctly regarding a hierarchical view concerning human anatomy and physiology, including a heterarchical model allows for an accurate whole body vision which embraces the contribution and distribution of forces throughout the body (Turvey and Fonseca, 2014). Hierarchy and heterarchy are both possible and permissible in biotensegrity.

Human movement is ultimately concerned with changes in shape in space over time. The concept of closed kinematic chains explains how movement in one body region (such as the foot) leads to movement at a distant body location (the opposite shoulder) (Findley et al., 2015).

Closed-chain kinematics of less than four bars result in no movement, or at best results in stable but rigid structures. The addition of a fourth and additional bar is necessitated to provoke and sustain controlled full-body motion. Kinematic chains, compiled of multiple bars, provide force couples which utilize combinations of three, four or multiple bar complexities. Such complexity provides energy efficient stability and motion throughout the entire biotensegrity (Levin et al., 2017).

Shedding light on fascia and biotensegrity in sport and movement

Sport scientists and movement specialists aim to provide training and movement strategies that

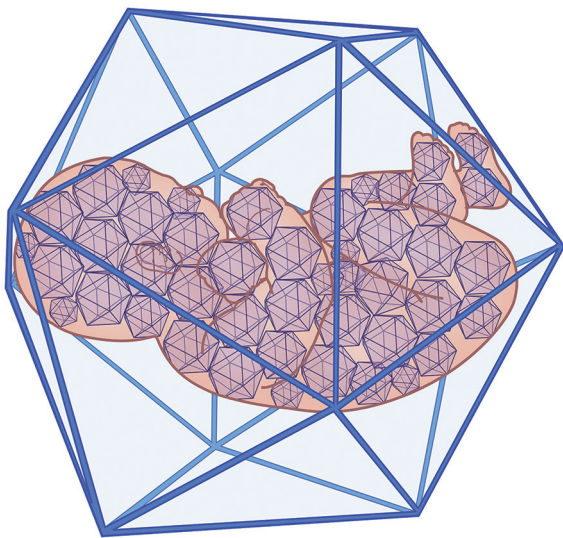


Figure 11.3

Icosahedron baby. Tensegrity within tensegrity within tensegrity. Our fractal nature (Sharkey, 2017)

Artist: Amanda Williams.

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improve performance maintain range of motion and reduce risk of injury. Unfortunately, the very nature of sport and the intense levels of training required, especially at an elite level, make this a difficult task. However, positive steps can be taken based on our current understanding of fascia-related science coupled with biotensegral principles to ensure better-informed choices and more appropriate recovery strategies.

Such strategies are applicable for the elite level athlete, the weekend warrior, or the less intense physical activities of members of all ages within our communities focusing their training on daily functionality and health-related benefits.

When we need to identify the root of a problem (e.g. an injury) we risk getting lost in the forest of “everything is connected to everything else”.

A torn calcaneal tendon will require localized attention and intervention to assist in healing.

The question that biotensegrity answers and the ethos that “everything is connected” raises is “What exactly is that connection?” Did the calcaneal injury occur as a result of a more pressing problem elsewhere in the body? Is the calcaneal injury simply the messenger and, if so, how do we locate the true source of the problem?

To begin to take a step closer to answering these questions, we must accept the first principle reality that all biotensegrities are non-linear.

This fact emphatically rules out linear approaches to movement, no matter how alluring. Linear-based fascial maps provide a model that is immediately simplistic in helping to bridge the reality of the fascia-muscle fiber continuity (Figure 11.4). While this model is appealing, it does not solve the issue of non-linearity. Training regimes need to be fascia science-based and biotensegrity-informed. Sport scientists must balance high-intensity physical routines,

prescribed to improve elite athletic performance, with appropriate recovery time to reduce the risk of fascial and connective tissue injury. Differences between biotensegrity-informed movement, to achieve goals which are sport-specific, as opposed to dealing with the functional activities required when dealing with a member of the public, need to be recognized.

It has been proposed that biotensegrity-based training principles may significantly reduce the risk of injury while increasing athletic performance (Kelsick, 2015).

The principle of specificity requires the need to repeat the specific movements or actions of any given sport, such as golf, rowing, tennis or javelin-throwing. In this regard, consideration for muscle fibers is critically important.

Figure 11.4 shows an image of deep fascial compartments of a human leg. This image, coupled with Figure 11.5, highlights the role of fascia



Figure 11.4 Shedding light on fascia and the biotensegrity model.

Image: Stecco, C., Sharkey, J. & Schleip, R. (2018). Human Fascial Net Plastination Project. Fascia Research Society

Credit: Thomas Stephan Photography.

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in the transfer of forces generated by muscle fibers, a process known as mechanotransduction (Ingber, 2008).

It is not muscle fibers that typically get damaged during intense exercise (Schleip, 2015) but rather the fascial tissue within which the muscle cells are embedded (Chapter 5).

Muscle fibers are an essential continuity of the myofascial and osseofascial systems. As there are only curves, helices and spirals in the human body, one must think in terms of 360° as the fascia enters, spirals, enfolds, penetrates, enmeshes and emerges from the various compartments. Fascia acts “to imprison its contents (vascular, neural, interstitial, lymphatic) and restrain, contain, separate, invaginate, and share forces generated within muscle fibers” (Sharkey, 2018a).

Forces generated in the fibers of any muscle are shared throughout the entire biotensegrity, locally and globally, via the softest and hardest fasciae. This changes our “one muscle, one movement” ideology. Motion is finely tuned

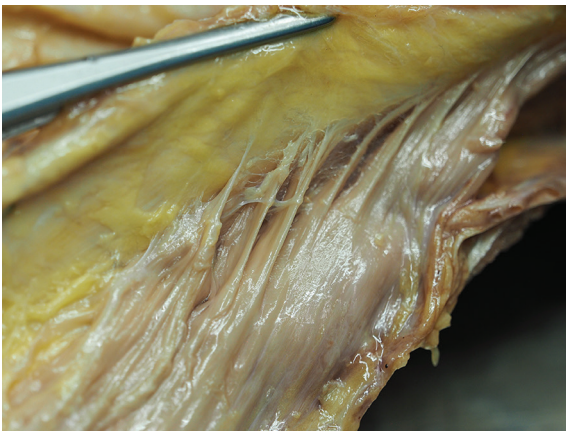


Figure 11.5

Beautifully capturing the continuity of muscle fibers within the fascia profunda.

Image: Stecco, C., Sharkey, J. & Schleip, R. (2018). Human Fascial Net Plastination Project. Fascia Research Society

through synergistic, multi-bar mechanisms facilitated by the multidimensional fascial fabric (Figures 11.4 and 11.6). This is comparable to a synchronized orchestra providing not one note, not one chord but rather a symphony of sound. Similar to sound, which travels in waves, muscle forces resonate and vibrate at specific electromagnetic frequencies shared through the osseofascial system, providing the cell-specific energy for metabolism, physiology and movement (Oschman, 2015).

Foams and froth require neighboring bubbles to move and adjust in response to mechanical



Figure 11.6

A historical image of the world's first 3D printed (mini-model version) of the deep fascial compartments of thigh. Sharkey, J. (2018).

Thanks to 3D Life Prints, NTC, Ireland, and the Fascia Research Society

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changes, reflecting a multi-bar, closed-chain organization.

Our true human fascial anatomy is more closely related to the froth on your morning coffee than it is to the mechanics of an automobile, a high-rise building, or any manmade, hard matter compression-based structure.

The language of biotensegrity is tension and compression with continuity. The premise

of biotensegrity is non-linear continuous matter that is self-generated (autopoiesis), self-organizing, self-stressing, hierarchical, load-distributing and low energy-consuming. There are no shear moments, no bending moments, no lever arm and no pin-joints. Sport science and movement therapies have entered an exciting new era of research with biotensegrity as the logical model underpinning future scientific investigation.

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Credit: special thanks to photographer Thomas Stephan (<https://thomas-stephan.com>).